Hack-Nocturne '26

Through the night Ideas take flight

Your Idea

Traditional cybersecurity reacts.
Modern phishing adapts in real-time.

PhishFree
Real-Time Multi-Modal AI Phishing Detection & Prevention

Problem

The Web is No Longer Human - It's AI vs AI

- Phishing has evolved into **AI-generated, hyper-personalized attacks** that bypass traditional filters.
- Zero-day domains and fast-flux infrastructure render **blacklist-based detection obsolete**.
- Attackers replicate **pixel-perfect login pages**, defeating text-only scanners.
- Users are forced to trust "red/green warnings" without understanding *why* a site is dangerous.
- Enterprises lack **tamper-proof forensic trails** for post-attack investigation.

Our Solution

Real-Time Multi-Modal Phishing Detection
A browser extension that analyzes every webpage using:
LLM → Detects social engineering & malicious intent
CNN → Identifies fake login page cloning
GNN → Maps suspicious domain networks

- Multi-model risk fusion for high accuracy
- Explainable threat breakdown for user trust
- Blockchain-backed tamper-proof verification logs

What Makes It Unique

- True multi-modal AI (Text + Visual + Domain Graph)
- Live detection before credential submission
- Explainable threat breakdown
- On-chain verification for forensic trust

Why should we select you?

We are highly interested in addressing real-world cybersecurity challenges, particularly in addressing phishing attacks caused by AI-based phishing attacks.

We are aware of how to design multi-layered systems, including how to incorporate AI models, backend APIs, browser integrations, and security considerations.

We are interested in building a scalable and interpretable phishing detection system, not just a classifier.

We are interested in exploring how multi-modal AI can greatly benefit the accuracy of phishing attacks.

This hackathon is a great opportunity to test our concept in a competitive environment.

We are highly interested in building a technically robust and feasible system.

Hack Nocturne gives us the perfect stage to transform this research-driven concept into a real-world cybersecurity solution.

Tech-Stack & Platforms You Mostly Build On

Frontend / Backend

Frontend
- HTML
- JavaScript
- Tailwind CSS
- Chrome Extension (Manifest V3)

Backend
- Python 3.10
- Flask REST API
- Flask-CORS
- Modular ML inference pipeline

AI & ML Stack

- PyTorch
- HuggingFace Transformers
- RoBERTa (Phishing language detection)
- CLIP + MLP Head (Visual spoof detection)
- GraphSAGE(Domain relationship modelling)
- PyTorch Geometric
- LightGBM (Ensemble risk scoring)

Cloud / DevOps / Tools

- Docker & Docker Compose
- GitHub
- REST API architecture
- Rate limiting & logging
- CSV report export

Blockchain & Web3 / Cybersecurity

- Wev3.py
- Ethereum Sepolia Testnet
- Immutable analysis anchoring
- Domain intelligence modelling
- Risk scoring engine

Leave fields **blank** where **not applicable**, you can put **logos** of tech stacks in the provided space

Thank you

Good Luck 👍

Team **Hack-Nocturne**

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Team – Fourage

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